Assessment 1: Project Proposal for a Machine Learning-based Solution

This assessment contributes to 30% of the final course grade

Assessment Overview

For this assessment, you will propose a project to design a machine learning-based solution for a real-world scenario. Your proposal will describe the significance of the real-world problem and propose a solution that applies appropriate and relevant machine-learning algorithms. You will include a methodology for developing your model and will outline how you plan to evaluate it through testing, responding to issues or challenges that arise.

Learning Outcomes

This assessment contributes to:

- **LO 1:** Analyse issues and challenges of machine learning algorithms for optimising applied industry solutions.
- **LO 2:** Evaluate the mathematical and statistical background of machine learning models to optimize solutions for industry.

Turnitin Canvas Plagiarism Framework

- This assignment will be checked for plagiarism and originality using the Turnitin Canvas Plagiarism Framework. You can learn more about this system by reading What is the Turnitin Canvas Plagiarism Framework.
  Before submitting your final version, you have the chance to upload drafts to Turnitin Draft Coach. To do this, read the sections What is Turnitin Draft Coach and How to use Turnitin Draft Coach and follow the instructions provided there for drafting your assignment.
- The similarity threshold is 15%.

Conditions

- All course materials, and any other resources, can be used to complete this assessment.
- The work you submit must be your own work. It is an individual assessment.
- You can ask a tutor to clarify the instructions, and/or for advice, but they cannot do/solve the required tasks – you must carry out all the tasks yourself!

Instructions

For this assessment, you are required to **pick a real-world problem** that requires a solution based on machine learning. You are strongly encouraged to find a problem related to an industry within New Zealand. You will write a proposal for a machine-learning project that will help solve this real-world problem.

- NOTE: Relevant datasets can be obtained from NZ data repositories, Kaggle
- Links to an external site.
- , or SemEval
- Links to an external site.
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- s to an external site.or **SemEval**.
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Structure of the Proposal

The proposal needs to be written using the following structure:

- Title page
- Table of Contents
- Introduction
    - Provide the selected problem's description in terms of wording and mathematical expressions (if any).
    - Justify the problem's significance (the **motivation** for the project)
    - Describe and analyse existing challenges in solving the problem.
    - Detail research objectives.
- Literature Review
    - Discuss background knowledge related to problem (what is the **context**?)
    - Analyse and evaluate approaches that have been tried in the past for this problem, if any (e.g., existing solutions/similar machine-learning products/ or state-of-the-art approaches).
    - Refer to minimum 2 – 3 research papers.
- Methodologies
    - Describe and justify the approach to be used to solve the problem.
    - Discuss the dataset to be used for the project.
    - Discuss the machine learning algorithms, data collection methods, feature engineering techniques etc. that you plan to use. Explain why they're suitable.
    - Clarify what you plan to do for data pre-processing.
    - Analyse the potential for issues and challenges to arise. How might they be mitigated?
- Evaluation

    - Discuss how the effectiveness of the solution will be measured and evaluated. What experiments are you planning to conduct?
    - Outline briefly what kind of results you expect.
    - How will you assess and respond to any predicted issues and challenges around applying machine learning algorithms to your solution?
- Conclusion
- References

The citations and references need to use the APA or IEEE format.

A template for the Project Proposal is available here, which you may choose to use. It provides a helpful model for how your proposal should look and what should be included.

Submission Checklist

Task	Evidence Required
1.	Real-world problem described and justified
2.	Background information / Existing Solutions / Similar machine-learning products/approaches evaluated
3.	Proposed methodology using machine-learning algorithms
4.	Dataset described
5.	Evaluation Plan for Testing and Experiments
6.	Issues and challenges predicted, responses proposed

Submission Instructions

Upload your document to Canvas, labelled "**StudentID_IT8105_Assess1**" in PDF, DOCX, DOC or ODT format.

Policies

Success Criteria

To be successful in each assessment you must obtain a minimum of 50% of the total available marks. You will be marked according to the rubrics shown in each assessment. Please take the time to study these as they contain information on how you will be assessed.

You are allowed a maximum of two attempts. The maximum percentage to be awarded on a second assessment attempt is 50%. Even if you score higher than 50% on a second attempt, you will only be awarded 50%.

Extensions and Late Submissions

An extension can only be granted under exceptional conditions normally for a maximum of (5 working days), unless there is evidence to support a longer timeframe. Complete and return the following form to your Course Lecturer:

Assessment Extension Request Form.pdf

Links to an external site.

Assessments received **one (1) to five (5)** working days past the due date, or an approved extended due date, will be marked but will receive a mark capped at the pass mark i.e. 50% for graded assessments.

Assessments received **more than five (5)** working days past the due date, or approved extended due date, will not be marked and will receive a mark of zero or fail.

Academic Integrity and Academic Misconduct Policy

Whitecliffe is committed to ensuring the highest level of academic integrity and expect staff and students to conduct themselves as responsible and honest members of the academic community.

Where there is evidence that breaches of academic integrity may have occurred, the course of Academic Integrity and Academic Misconduct Policy

Links to an external site.

will be followed.

This policy considers academic integrity to mean honesty in all matters relating to academic endeavours. Breaches in academic integrity, or academic misconduct, include all forms of cheating incorporates plagiarism (intentional or unintentional), unauthorised collaboration, impersonation, falsification and collusion.

Declaration

By submitting your work, you are indicating that you agree to the following declaration:

"The work presented in this assessment is to the best of my knowledge original, except as acknowledged in the text, and the material has not already been submitted, either in whole or in part, for any academic award at this or any other tertiary institution. I promise not to share this assessment in part or whole with any other student at Whitecliffe or outside this campus."

Rubric

IT8105 Assessment 1: Project Proposal for Machine Learning-based Solution

Criteria	Ratings	Pts
This criterion is linked to a Learning Outcome Real-World Problem: Relevance and necessity for a machine-learnin	**15 to >13.0 ptsConvincingly justifies the significance of the dynamic, real-world problem and its need for a machine-learning solution. Research objectives are precisely stated and linked to the problem.** **13 to >11.0 ptsSomewhat justifies the significance of a relevant, real-world problem and its need for a machine-learning solution. Research objectives are stated but lack precision or connection to the problem.** **11 to >8.0 ptsDescribes but does not fully justify the significance of a relevant, real-world problem and its need for a machine-learning solution. Research objectives are mentioned but lack detail and are not reasonable.** **8 to >0 ptsLittle attempt to argue the significance of the selected, real-world problem / or the real-world problem has no relevant need for a machine-learning solution. Research objectives are vaguely stated or missing.**	15 pts

g solution		
This criterion is linked to a Learning Outcome Context: Background information, existing solutions/similar machine-learning products/ or state-of-the-art approaches.	**15 to >13.0 ptsAnalyses in detail useful background information. Provides a detailed description and critical evaluation of several relevant existing solutions or similar products / approaches.** **13 to >11.0 ptsAnalyses with clarity some relevant background information. Provides a clear description and moderate evaluation of several relevant existing solutions or similar products / approaches.** **11 to >8.0 ptsIncludes limited, relevant background information. Describes one or two relevant existing solutions or similar products / approaches, and/or minimal evaluation is provided.** **8 to >0 ptsMinimal background information provided. Little to no attempt to describe relevant existing solutions or similar products / approaches, and/or examples provided lack relevance.**	15 pts

This criterion is linked to a Learning Outcome Proposed Solution: Relevance & Potential	**20 to >16.0 ptsProposes a detailed solution using highly relevant and appropriate machine-learning algorithms. Solution demonstrates excellent potential to solve the real-world problem.** **16 to >13.0 ptsProposes a solution using mostly relevant and appropriate machine-learning algorithms. Solution demonstrates reasonable potential to solve the real-world problem.** **13 to >10.0 ptsProposes a solution using basic machine-learning algorithms that are mostly appropriate. Solution demonstrates limited potential to solve the real-world problem.** **10 to >0 ptsProposes a solution using inappropriate, irrelevant or insufficient machine-learning algorithms. Solution demonstrates little to no potential to solve the real-world problem.**	20 pts
This criterion is linked to a Learning Outcome Proposed Solution: Methodologies	**20 to >16.0 ptsOutlines detailed, sophisticated, and highly appropriate methodologies for implementing the design of the solution.** **16 to >13.0 ptsOutlines clearly a relevant and appropriate methodologies for implementing the design of the solution.** **13 to >10.0 ptsDescribes, with limited clarity, mostly appropriate methodologyies for implementing the design of the solution.** **10 to >0 ptsLittle to no attempt to describe a methodology for implementing the design of the solution, and/or methodology is inappropriate.**	20 pts

This criterion is linked to a Learning Outcome Solution Evaluation: Planned experiments and predicted issues and challenges	**30 to >24.0 ptsOutlines a detailed plan for thorough and highly appropriate solution testing. Insightful and resourceful responses proposed for predicted issues / challenges.** **24 to >20.0 ptsOutlines a clear plan for appropriate solution testing. Reasonable responses proposed for predicted issues / challenges.** **20 to >15.0 ptsDescribes, with limited clarity, a plan for mostly appropriate solution testing. Limited responses are proposed for one or two predicted issues / challenges.** **15 to >0 ptsLittle to no attempt to plan relevant solution testing or predict issues or challenges.**	30 pts

Total Points: 100

Assignment topic:
Hi Samina,

Thank you for writing to me about the proposed topic change. I have to be direct: I do not approve the change. The new topic is weaker than your original one, and I would like you to stay with "Area-Level Crime Hotspot and High-Victimisation Risk Prediction", or at least a refined version of it.

My discussion is below.

Sentiment analysis and complaint classification on restaurant reviews is the "Hello World" of NLP. GeeksforGeeks publishes it as a beginner tutorial exercise. A lot of

other tutorials and papers doing the same thing published in previous years. Some links below:

https://www.geeksforgeeks.org/nlp/sentiment-analysis-of-restaurant-reviews/
https://ieeexplore.ieee.org/document/9300098
https://www.sciencedirect.com/science/article/pii/S1877050924009876
https://medium.com/@vincents0511/sentiment-analysis-of-restaurant-reviews-430875614095

This topic is as saturated. There is nothing left to contribute by replicating it, and it does not meet master's-level standards.

Your proposal also has no dataset, no methodology, and the phrase "prioritise issues based on their importance or frequency" describes counting and sorting, which is descriptive statistics, not machine learning.

Your original topic is substantially stronger. Area-level crime prediction using spatial and demographic features is a genuine ML problem with policy relevance, and New Zealand has excellent public data to support it. NZ Police publishes detailed crime statistics on policedata.nz (https://www.police.govt.nz/about-us/publications-statistics/data-and-statistics/policedatanz), updated monthly, including:

- Victimisation Time and Place: where and when crimes were committed, broken down by geographic unit (meshblock and area unit), offence type, and time period
- Victimisation Demographics: age, sex, and ethnicity of victims by police district
- Proceedings/Offender Demographics: offender data by offence type and police area

This data is available from 2014 onwards and is downloadable directly from the website. There is also a Kaggle dataset ("NZ Police Victimisation Time and Place Dataset") covering 2014–2021 if you prefer a pre-packaged format, though I would recommend going to the primary source for the most current data.

The NZ Crime and Victims Survey (NZCVS), run by the Ministry of Justice, provides additional survey-based victimisation data including unreported crime, with results published annually.

A project predicting area-level victimisation risk from spatial, temporal, and demographic features using the policedata.nz datasets combined with area-level census data from Stats NZ would be a more genuine contribution. The methodological challenges are real: spatial autocorrelation must be handled, temporal validation is needed, class imbalance may arise if you define "hotspot" as a

binary outcome, and interpretability (SHAP) is essential because the audience is police and policymakers who need to understand why an area is flagged.

I would encourage you to go back to your original topic, download the policedata.nz data, and develop a proper proposal around it.

In case you want to discuss more, please do not hesitate to contact me.

Regards,

Dr Joseph Dang

Senior Lecturer | Technology